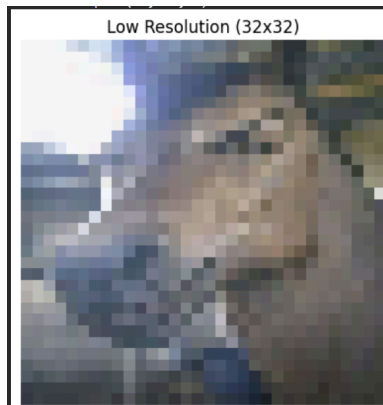


Method

The image was first reduced to a lower resolution to make the differences more visible. Then, three down sampling methods (max, average, and median) were applied. After that, the image was upsampled using nearest neighbor, bilinear, and bicubic interpolation.

```
# =====  
# FORCE LOW RESOLUTION  
# =====  
low_res = cv2.resize(img, (32, 32))  
  
print("Low-res shape:", low_res.shape)  
  
plt.imshow(low_res)  
plt.title("Low Resolution (32x32)")  
plt.axis("off")  
plt.show()
```

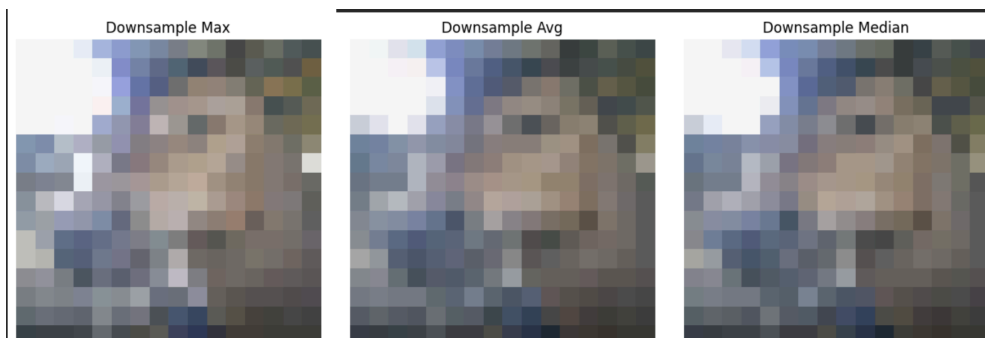


Down Sampling Results

Max down sampling keeps the highest pixel value, so the image looks a bit sharper but can be too strong in bright areas.

Average down sampling makes the image smoother but also more blurry because it averages pixel values.

Median down sampling reduces noise and keeps edges better compared to average, so it looks more balanced.



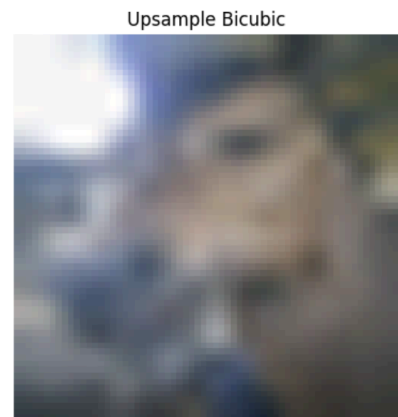
Up Sampling Results

Nearest neighbor gives a pixelated result because it just copies nearby pixels.

Bilinear interpolation is smoother but slightly blurry.

Bicubic interpolation produces the best result because it keeps smoother edges and more detail.

```
# =====  
# UPSAMPLING (FROM DOWNSAMPLED)  
# =====  
def upsample(image, scale=4):  
    nn = cv2.resize(image, None, fx=scale, fy=scale, interpolation=cv2.INTER_NEAREST)  
    bilinear = cv2.resize(image, None, fx=scale, fy=scale, interpolation=cv2.INTER_LINEAR)  
    bicubic = cv2.resize(image, None, fx=scale, fy=scale, interpolation=cv2.INTER_CUBIC)  
    return nn, bilinear, bicubic  
  
us_nn, us_bilinear, us_bicubic = upsample(ds_avg, 4)
```



Observation

The differences between methods are not very obvious when using high-resolution images. After reducing the image size, the effects of each method become clearer.

Conclusion

Different sampling methods produce different visual results. Max highlights intensity, average smooths the image, and median preserves edges better. For up sampling, bicubic gives the best quality, while nearest neighbor is the simplest but lowest quality.